Multi-Role Buffer Protocol

Batch Manufacturing Record

Run: Run-2026-001 Status: ACTIVE Started: Completed:

Project: Protocol Version: v2 Generated: April 1, 2026

1. General Information

Product Name / Candidate	Multi-Role Buffer Protocol
Batch / Lot Number	Run-2026-001
Target Yield / Scale	
Process SOP / Protocol Reference	Multi-Role Buffer Protocol (v2)

2. Bill of Materials (BOM)

Record all raw materials, media, and buffers utilized in this run.

Material Description	Part / Lot Number	Required Quantity	Actual Quantity Added	Operator Initials & Date

3. Equipment Log

Record primary equipment used to ensure traceability.

Equipment Name / Type	Equipment ID (Asset Tag)	Calibration Status (Valid Until)	Operator Initials & Date

4. Execution: Unit Operations

Execute process steps. Record actual values, operator, and verifier (if required by SOP) for each step.

4.1 Media Prep

Step	Instruction	Target Parameter	Actual Value	Operator	Verifier
1	Weigh Reagents	Weigh NaCl and KCl. NaCl: 8, and KCl: 0.2.	**NaCl (g)** 8.01 **KCl (g)** 0.2	D.S.C.	
2	Dissolve	Add to 1000 mL water and stir.	1000	D.S.C.	

4.2 QC

Step	Instruction	Target Parameter	Actual Value	Operator	Verifier
1	Measure pH	Measure and adjust pH. Target pH: 7.4.	7.38	J.W.	

5. Deviations and Process Comments

Log any deviations from target parameters or SOP. Run notes and anomalies are surfaced here.

Step / Status Ref.	Description of Deviation / Observation	Impact Assessment Required? (Y/N)	Lead Reviewer Sign-off
ACTIVE	Observed slight turbidity after buffer prep.		Dr. Sarah Chen 2026-04-01 10:35:00
ACTIVE	[ANOMALY] ANOMALY: Cell viability below threshold at 85%.		Dr. Sarah Chen 2026-04-01 11:20:00

6. Final Disposition & Signatures

By signing below, the responsible parties certify that the material described herein was produced according to the specified process and that all deviations have been reviewed.

Wet-Ink Sign-Off

Role	Name (Print)	Signature	Date
Lead Operator / Engineer			
Process Development Lead			
Quality / Compliance (if applicable)			